

DESIGN PATTERNS

Subject Code: A70530

Regulations: R15 – JNTUH

Class: IV Year B.Tech CSE I Semester



Department of Computer Science and Engineering

BHARAT INSTITUTE OF ENGINEERING AND TECHNOLOGY

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DESIGN PATTERNS (A70530)

COURSE PLANNER

I.COURSE OVERVIEW:

The course covers a wide range of software development concepts, abilities, and skills, from analyzing a problem to implementing a solution, also discuss the design patterns in Smalltalk MVC architecture, Express representation invariants, understand their impact on efficiency and ease of implementation, and implement them as runtime assertions. Outlines the differences between structural patterns and behavioral patterns of a model. The course explains about common design vocabulary. This course helps to determine how to be recognizing a design and they can reduce the amount of refactoring, helps to use primitive techniques such as objects, inheritance, and polymorphism. Describes problems that occur in a design how to resolve them and how to evaluate them.

II.PRE-REQUISITES:

- Object Oriented Analysis and Design.
- Data structures and algorithms.
- Programming Language (C++ or Java)

III.COURSE OBJECTIVE:

1	Demonstration of patterns related to object oriented design.
2	Describe the design patterns that are common in software applications.
3	Analyze a software development problem and express it.
4	Design a module structure to solve a problem, and evaluate alternatives.
5	Implement a module so that it executes efficiently and correctly.

IV.COURSE OUTCOMES:

S.NO	Description	Bloom's Taxonomy Level
1	Construct a design consisting of a collection of modules.	Create
2	Exploit well-known design patterns (such as Iterator, Observer, Factory and Visitor).	Analyze
3	Distinguish between different categories of design patterns.	Analyze
4	Ability to understand and apply common design patterns to incremental/iterative development.	Understand
5	Ability to identify appropriate patterns for design of given problem.	Remember
6	Design the software using Pattern Oriented Architectures.	Create

V.HOW PROGRAM OUTCOMES ARE ASSESSED:

Program Outcomes		Level	Proficiency assessed by
PO1:	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.	3	Assignment, Exercises
PO2:	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.	2	Exercises
PO3:	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	2	Exercises

PO4:	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	--	
PO5:	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.	2	Design, Exercises
PO6:	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.	--	
PO7:	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.	1	Assignment, Exercises
PO8:	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.	1	Seminars, Discussions
PO9:	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary teams.	1	Workshop
PO10:	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	1	Seminars, Paper presentations
PO11:	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	1	Design Exercises, Discussions
PO12:	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	1	Exams, Discussions

VI.HOW PROGRAM SPECIFIC OUTCOMES ARE ASSESSED:

PROGRAM SPECIFIC OUTCOMES		Level	Proficiency assessed by
PSO1:	To use mathematical methodologies to crack problem using suitable mathematical analysis, data structure and suitable algorithm.	2	Lectures, Assignments
PSO2:	The ability to interpret the fundamental concepts and methodology of computer systems. Students can understand the functionality of hardware and software aspects of computer systems.	3	Projects
PSO3:	The ability to grasp the software development lifecycle and methodologies of software systems. Possess competent skills and knowledge of software design process. Familiarity and practical proficiency with a broad area of programming concepts and provide new ideas and innovations towards research	2	Guest Lectures

VII.COURSE CONTENTS – AS PER JNTUH SYLLABUS:

UNIT – I:

What is a Design Pattern?, Design Patterns in Smalltalk MVC, Describing Design Patterns, The Catalogue of Design Patterns, Organizing The Catalog, How Design Patterns solve Design Problems, How to Select a Design pattern, How to Use a Design Pattern.

.UNIT – II

A Case Study: Designing a Document Editor, Design Problems , Document Structure, Formatting , Embellishing the User Interface, Supporting Multiple Look-and-Feel Standards, Supporting Multiple Window Systems, User Operations Spelling Checking and Hyphenation, Summary, Creational Patterns, Abstract Factory, Builder , Factory Method, Prototype, Singleton, Discussion of Creational Patterns.

UNIT – III:

Structural Pattern Part-I, Adapter, Bridge, Composite.

Structural Pattern Part-II, Decorator, Facade, Flyweight, Proxy.

UNIT--IV:

Behavioral Patterns Part: I, Chain of Responsibility, Command, Interpreter, Iterator.

Behavioral Patterns Part: II, Mediator, Memento, Observer, Discussion of Behavioral Patterns.

UNIT V:

Behavioral Patterns Part: III, State, Strategy, Template Method, Visitor, Discussion of Behavioral Patterns. What to Expect from Design Patterns, A Brief History, The Pattern Community, An Invitation, A Parting Thought.

RELEVANT SYLLABUS FOR GATE: NA

RELEVANT SYLLABUS FOR IES: NA

LESSON PLAN:

LESSON PLAN:

Lecture no	Week no	Topic to be covered	Learning Objectives	References
		Unit I		
1	1	What is a Design Pattern?	Recognize the importance of design and design patterns.	1. Design Patterns By Erich Gamma, Pearson Education. 2. Patterns in JAVA Volume -I By Mark Grand, Wiley Dream
2		Design Patterns in Smalltalk MVC		
3		Describing Design Patterns		
4		The Catalog of Design Patterns		
5	2	The Catalog of Design Patterns	Listing contents of catalog of design patterns.	
6		Organizing The Catalog		
7		How Design Patterns solve Design, Problems	Determine object granularity and Specify object interfaces.	
8		How to Select a Design pattern		
9	3	How to Use a Design Pattern	Outlines the problems of Lexi's design.	
10		Revision of Unit-1 / Tutorial Class		
Unit II				
11		A Case Study: Designing a Document Editor		
12		Design Problems		
13		Document Structure		
14	4	Formatting		

15		Embellishing the User Interface.		Tech.
16		Supporting Multiple Look-and-Feel Standards		
14		Supporting Multiple Window Systems	Summarize command class and subclasses.	1. Design Patterns By Erich Gamma, Pearson Education.
15	5	User Operations Spelling Checking and Hyphenation.	Describe interact or class and subclasses.	2. Patterns in JAVA Volume -I By Mark Grand, Wiley Dream Tech.
16		Summary		
17		Creational Patterns		
19		Factory Method		
20	6	Abstract Factory	Lists out the importance of creational patterns.	
21		Builder		
22		Prototype		
23		Singleton	Reviews the importance of prototype and singleton pattern.	
24		Discussion of Creational Patterns		
25		Revision of Unit-2 / Tutorial Class		
26	7	Unit III Structural Pattern Part-I	Identify the importance of structural patterns.	
27		Adapter		1. Design Patterns By Erich Gamma, Pearson Education.
28	8	Bridge		
29		composite		
30		Mock Test -1		
31		Bridge Class		
32	9	Structural Pattern Part-II	Summarizes the implementation issues when applying the bridge pattern.	
33		Decorator		
34		Facade		
35		Flyweight,proxy		
36	10	Mid1		
37		UNIT-IV Behavioural Patterns Part: I		2. Patterns in JAVA Volume -I By Mark Grand, Wiley Dream Tech
38		Chain of Responsibility		
39		Command	Outlines the importance of behavioral patterns.	
40		Interpreter		
41	11	Iterator		
42		Behavioral Patterns Part: II		
43		Mediator		
44	12	Memento		
45		Observer	Use of observer, Mediator, Memento in	
46		Discussion of Behavioral Patterns		

47		Discussion of Behavioral Patterns	behavioral patterns.	
48	13	UNIT V		1. Design Patterns By Erich Gamma, Pearson Education. 2. Patterns in JAVA Volume -I By Mark Grand, Wiley Dream Tech education.
49		Behavioral Patterns Part: III	Use of state, strategy in behavioral patterns.	
50		State		
51		Strategy		
52	14	Template Method	Identify the need of template method and visitor Discussion of Behavioral Patterns.	
53		Visitor		
54		Discussion of Behavioral Patterns	Summarizes the importance of design patterns.	
55		What to Expect from Design Patterns		
56	15	A Brief History	Reviews the template of design patterns.	
57		The Pattern Community		
58		An Invitation		
59		A Parting Thought		
60	16	Mock Test - 2		
61		Bridge Class		
62		Revision of Unit-5		
63		Tutorial Class		
64		Revision of Syllabus		

MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course outcomes	Program outcomes												PROGRAM SPECIFIC OUTCOMES		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
1	3	--	2	--	--	--	--	--	--	--	--	--	2	--	--
2	3	2	--	--	--	--	--	--	--	--	--	--	--	--	2
3	3	--	2	2	2	--	--	--	--	--	--	--	2	--	--
4	--	2	2	--	--	--	--	--	--	--	--	--	--	3	--
5	--	2	2	--	--	--	--	--	--	--	--	--	--	--	--
6	3	--	2	--	2	--	--	--	--	--	--	--	2	--	--

S=Supportive

H=Highly Related

SUGGESTED BOOKS:

A. Text books:

T1: Design Patterns By Erich Gamma, Pearson Education

REFERENCE BOOKS:

R1.Patterns in JAVA Vol-I (or) Vol-II By Mark Grand, Wiley Dream Tech.

R2.Java Enterprise Design Patterns Vol-III By Mark Grand Wiley Dream Tech**QUESTION BANK:****A. DESCRIPTIVE:****UNIT – 1****SHORT ANSWER QUESTIONS**

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	Define design pattern	Remember	1
2	List the four elements of design patterns.	Knowledge	1
3	Sketch MVC architecture	Knowledge	3
4	Write the different sections of design patterns	Knowledge	1
5	Give an example for class structure	Understand	2

LONGANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcome
1	Discuss the MVC architecture in small talk	Understand	1
2	List the different sections of design patterns. Explain about them	Knowledge	1
3	State and explain the classification of design patterns.	Knowledge	2
4	Name the patterns along with their intents that are included in the catalog of design patterns.	Knowledge	2
5	List the various ways of organizing the design patterns.	Knowledge	2

UNIT -2**SHORT ANSWER QUESTIONS**

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	a) Give the meaning of “WYSIWYG” b) Define an Iterator class c) Define creational design patterns.	Understand , Remember	1,2
2	Define a) Lexi. b) Formatting. C) Abstract class.	Remember	3,1,3
3	Differentiate compositor and composition	Knowledge	2
4	List five types of creational patterns	Knowledge	2
5	Define structure and applicability of abstract factory method.	Remember	3

LONGANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	a) Explain partial Glyph class hierarchy with a neat diagram. b) Explain in detail about the 'Glyph' abstract class.	Understand	1,3
2	a) Explain the implementation of abstract factory. b) Discuss about structure, applicability, participants of abstract factory method	Understand	3
3	a) Discuss the two embellishments that can be added to Lexi's user interface. Signify the role of decorator pattern in embellishment. b) Explain about abstracting object creation in detail. .	Understand	2
4	a) Explain the structure of builder design pattern with class diagram and describe the collaboration with a sequence diagram. b) Give the sample code for abstract factory design pattern.	Understand	3,2
5	a) Explain the process of recursive composition in building a document. b) Discuss the implementation issues of builder design pattern.	Understand	3

UNIT – 3

SHORT ANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	Define structural patterns	Remember	1
2	Define a) composite design pattern b) bridge pattern	Remember	3,2
3	Sketch the structure of decorator pattern and List the use of decorator pattern.	Knowledge	4
4	Sketch the structure of Façade pattern	Remember	5
5	Define Intent and Consequences of bridge pattern	Remember	3

LONGANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	a) Describe the intent, motivation and applications of composite patterns. b) Discuss the importance of implementation in composite structural pattern.	Knowledge, Understand	2,2
2	a) Explain composite pattern with an example from drawing editor. b) Describe the consequences of composite design pattern.	Understand, Knowledge	3,3
3	a) Explain the structure of composite pattern with a class diagram b) Explain the structure of decorator pattern with suitable	Understand	3,4

	example		
4	a) Explain the motivation for decorator method with relevant patterns. b) Explain the structure of decorator design pattern with a class diagram.	Understand	4
5	a) Explain the structure of façade design pattern with class diagram and consequences. b) Explain the consequences of flyweight method with its structure.	Understand	5,4

UNIT -4

SHORT ANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	Define behavioural class pattern and Define behavioural object pattern.	Remember	4,5
2	Define a) Façade pattern b) decorator pattern. c) command pattern	Remember	5,3,5
3	Sketch sequence diagram for any example pattern.	Knowledge	5
4	Sketch the structure of server in election commission.	Knowledge	6
5	List the benefits of decorator pattern.	Knowledge	4

LONGANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	Explain the uses of command design pattern and its structure with class diagram. Also explain the implementation issues.	Understand	6
2	Explain the structure of façade design pattern with class diagram and consequences.	Understand	5
3	a) Explain the consequences of flyweight method with its structure. b) Explain structure of flyweight pattern with suitable example.	Understand	5
4	Explain the interpreter design pattern and discuss the consequences and implementation issues.	Understand	5
5	a) Explain the collaborations and consequences of command method. b) Explain the implementation issues of command pattern with sample code.	Understand	6,4

Unit-5

SHORT ANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
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1	Define a) mediator pattern b) memento pattern.	Remember	4,5
2	Sketch the class diagram for TCP connection class.	Knowledge	6
3	List the situations where state pattern can be used.	Knowledge	6
4	Define the phrase “objects for states”.	Remember	5
5	Define a) table driven approach b) state pattern.	Remember	6,5

LONGANSWER QUESTIONS

S.N O	QUESTION	Blooms Taxonomy Level	Course Outcom e
1	Explain the structure of mediator design pattern with a class diagram and discuss the collaboration with a sequence diagram.	Understand	5
2	Discuss about intent, motivation, structure, applicability and consequences of a mediator behavioral pattern.	Understand	5
3	Discuss the several ways the design pattern affect the way object-oriented software is designed.	Understand	5
4	a) Write about the two ways of grouping the patterns according to Christopher Alexander. b) Differentiate Alexander’s patterns and Design patterns.	Knowledge	6,6
5	a) Explain the implementation issues of observer design pattern. b) Discuss the history of design patterns.	Understand	6,6

B. OBJECTIVE:

Unit: I

1. MVC consists of 3 kinds of objects _____, _____, _____
2. The 4 essential elements in a design patterns are _____, _____, _____ and _____
3. A view object in MVC is the _____ object.
4. Objects are created by _____ a Class
5. Often an application will incorporate Classes from one or more libraries of predefined classes known as _____

UNIT-II

1. A _____ Abstract class for all objects that can appear in the Document structure
2. A _____ emphasis is on families of product objects
3. _____ is also Known as virtual constructor.
4. A _____ is a class of class and it has one instance.
5. We use an abstract class called _____ to define a general interface for access and traversal

UNIT-III

1. The intent of _____ Pattern is to compose the objects into tree structures to represent part-whole hierarchies.
2. _____ Call operation on an Adapter instance.
3. Decorator Forwards requests to its _____ Object.
4. We use Facade Pattern if you want to provide a _____ Interface to a _____ subsystem.

5. A proxy is used to provide a _____ for another object to control access to it.

UNIT-IV

1. Interpreter is used when efficiency is not a _____
2. A memento pattern is applied when a direct interface to obtaining the state would expose _____ details and break the object's encapsulation.
3. We use Iterator pattern to support _____ of aggregate objects.
4. A _____ Pattern is used to define an object that encapsulates how a set of objects interact.
5. The interpreter pattern describes how to define a grammar for _____.

UNIT-V

1. _____ Provides a common vocabulary for designers to use to communicate, document, and explore design alternatives
2. One of the Problem in developing reusable software is that it often has to be ____ or ____.
3. Brian Foote identified the three phases of life cycle of Object oriented software, those are _____, _____, and _____.
4. Design Patterns are especially useful in turning an _____ model into _____ model.
5. Design patterns show how to use primitive techniques such as _____ , _____ and _____.

WEB SITES ADDRESSES:

1. <http://www.oodesign.com/>
2. <http://ui-patterns.com/patterns/>
3. http://sourcemaking.com/design_patterns
4. http://www.dmoz.org/computers/programming/methodologies/patterns_and_anti-patterns/

EXPERT DETAILS:

- (1) Erich Gamma, University of Zurich
- (2) Martin flower, martinfowler.com
- (3) Kent Beck, University of Oregon

JOURNALS:

<http://www.emeraldinsight.com/>

http://auspace.athabascau.ca/bitstream/2149/742/1/design_patterns.pdf

LIST OF TOPICS FOR STUDENT SEMINARS:

1. Design pattern visualization for program Comprehension
2. The concept of Patterns
3. Problems in designing an application
4. How to support multiple look and feel standards
5. Significance of Abstract Pattern, Factory Method
6. Significance of Adapter Pattern, Bridge Pattern
7. Significance of Decorator Pattern, Façade Pattern
8. Significance of Command Pattern, Proxy Pattern
9. Significance of Mediator Pattern, Template Pattern

CASE STUDIES:

1. Case Study: Expression Tree Evaluator
2. Design Patterns in the Expression Tree Program
3. Design Patterns in the System Sort
4. Using the Factory Method Pattern for Access Table Initialization